

A Novel Approach to Reducing Radiation Dose in PET Imaging Through Magnetic Field Manipulation of Positronium

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1 Introduction

There is a plethora of imaging techniques at a patient’s disposal to improve their quality of treatment. Positron emission tomography (PET) in particular is an invaluable tool in modern medicine because it allows clinicians to visualize biological function rather than anatomy alone then permitted in magnetic resonance imaging (MRI) and computed tomography (CT). Critically, diseases are able to now be detected before structural changes become apparent since this technology assists in the visualization of metabolic activity (Surti, 2015; Vandenberghe et al., 2016). This is achieved through the intravenous injection of a positron-emitting radiotracer, fluorine-18 fluorodeoxyglucose (^{18}F -FDG), in a patient. After some period of time—traveling a short distance and thermalizing in surrounding tissue—the emitted positron annihilates with an electron, producing photons that are detected in temporal coincidence by a ring of scintillation detectors. These coincidence events define lines of response that are subsequently reconstructed into three-dimensional images of radiotracer distribution (Surti, 2015).

Despite advances in time-of-flight reconstruction and iterative image reconstruction algorithms, PET image quality remains fundamentally constrained by noise. Clinical protocols frequently attempt to increase the number of detected coincidence events by increasing the administered radiotracer activity in order to improve the noise to signal ratio. This improvement, however, comes at the cost of greater radiation exposure to the patient. The safety of the patient is of utmost priority, thus reducing injected activity while preserving diagnostic image quality has become one of the largest goals in contemporary PET instrumentation and image reconstruction research (Lois et al., 2010; Huang et al., 2009).

In computed tomography, X-ray attenuation itself constitutes the measured signal used to reconstruct tissue density. In PET however, photon interactions that prevent coincidence detection contribute radiation dose without providing useful imaging information (Cherry et al., 2012). Thus, an important objective in PET instrumentation is increasing the proportion of radioactive decay events that generate reconstructable coincidence signals while minimizing unnecessary radiation exposure (Lois et al., 2010; Surti, 2015).

The development of hybrid PET/MRI systems has created an opportunity to reconsider this problem from a quantum-mechanical perspective. Specifically, PET/MRI combines the molecular sensitivity of PET with the superior soft-tissue contrast of MRI while eliminating the additional radiation associated with CT acquisition (Antoch & Bockisch, 2009). More importantly, PET/MRI places positron annihilation events within strong static magnetic fields. Previous studies have primarily considered these magnetic fields as engineering constraints affecting positron range, and image reconstruction (Judenhofer et al., 2008; Partovi et al., 2014). This notion is not entirely solid as magnetic fields also interact directly with the quantum states of electron-positron systems, suggesting that they may influence the annihilation process itself.

The magnetic field interacts with the unstable bound state of electron and positron (before annihilation) called positronium. Positronium exists in two spin states, para-positronium (p-Ps) and Ortho-positronium (o-Ps). Para-positronium (p-Ps) is the spin-singlet state with $S = 0$ and it decays predominantly to two photons (two 511-keV gamma rays) which are used in conventional PET for image reconstruction. Ortho-positronium (o-Ps) is the spin-triplet state with $S = 1$ and it decays predominantly to three photons in vacuum due to the

charge-conjugation symmetry. In the biological environment, interactions such as pick-off annihilation and spin exchange reduce the lifetime of o-Ps considerably and thus the behavior of positronium is very sensitive to the surrounding molecular environment (Moskal et al., 2019; (Moskal et al., 2021)).

The quantum-mechanical properties of positronium have been extensively studied for more than half a century. External magnetic fields produce Zeeman splitting of the triplet manifold and induce mixing between the para-positronium state and the $m = 0$ ortho-positronium sublevel, thereby modifying annihilation probabilities through hyperfine interactions (Hughes et al., 1957; Namba, 2012). More recent positron annihilation lifetime and Doppler broadening measurements have experimentally demonstrated magnetic quenching of positronium in condensed matter, confirming that applied magnetic fields measurably alter positronium spin dynamics and annihilation behavior (Liu et al., 2020). These results establish that positronium annihilation is not solely determined by radioactive decay, but is also governed by quantum spin interactions that may be externally manipulated.

Recently, Wang and Huq (2023) proposed that this phenomenon may have direct implications for PET imaging. They argued that magnetic-field-induced hyperfine splitting within PET/MRI environments can alter the relative populations of para- and ortho-positronium, thereby modifying the ratio of two-photon to three-photon annihilation events. Because conventional PET reconstruction relies almost exclusively on two-photon coincidence detection, increasing the effective fraction of two-photon annihilations could improve PET counting efficiency without increasing administered radiotracer activity. Conversely, they demonstrated that resonant microwave radiation near the positronium hyperfine transition could induce the opposite effect, motivating new shielding considerations for PET/MRI facilities (Wang & Huq, 2023).

This paper builds upon this emerging concept by developing a rigorous quantum-mechanical framework describing magnetic manipulation of positronium spin states for PET imaging. We seek to investigate whether positronium’s spin-state populations may be intentionally engineered through externally applied magnetic fields. Using the formalism of spin operators, hyperfine interactions, Zeeman perturbation theory, and positronium eigenstates, we examine how magnetic-field-induced spin mixing influences annihilation probabilities and evaluate the potential consequences for PET signal generation. If the proportion of reconstructable two-photon annihilation events can be increased through quantum control of positronium spin states, PET image quality may be maintained while reducing the administered ^{18}F -FDG activity required for diagnostic imaging.

Ultimately, this paper explores a broader question that lies at the intersection of quantum mechanics and medical physics: whether quantum control of fundamental particle interactions can be translated into clinically meaningful improvements in molecular imaging. This approach proposed here seeks to improve PET performance by increasing the efficiency with which radioactive decay events become useful imaging information.

2 Methods

Physics of Zeeman mixing and Magnetic Quenching

Quantum States of positronium without external Magnetic field

The Hamiltonian of Positronium at ground state ($L = 0$) has only 1 term for spin-spin interaction: $H = s_p * s_e$, note that the spin is a vector. For convenience, let $s_1 = s_p$ and $s_2 = s_e$ so we re-write the Hamiltonian

$$H = A(s_1 * s_2)$$

in which $A > 0$, is so-called hyper-split factor.

Here we play some mathematical tricks: $S = s_1 + s_2$ (all vectors).

$$S^2 = (s_1 + s_2) * (s_1 + s_2) = s_1^2 + s_2^2 + 2(s_1 * s_2)$$

Then

$$s_1 * s_2 = \frac{S^2 - s_1^2 - s_2^2}{2}$$

(remember, they are still all vectors)

By quantum mechanics, operator s^2 acts on wavefunction and have eigen value $s(s + 1)$

If we replace those vectors above by the numbers, we have:

Then

$$s_1 * s_2 = \frac{S(S + 1) - s_1(s_1 + 1) - s_2(s_2 + 1)}{2}$$

We know $s_1 = s_2 = \frac{1}{2}$ (here set $\hbar = 1$)

Therefore

$$s_1 * s_2 = \frac{S(S + 1)}{2} - \frac{3}{4}$$

When we put 2 spins together, S is either 1 (aligned) or 0 (anti-aligned), therefore $S = 1$, or 0.

For $S = 1$ triplets, $H = A(s_1 * s_2) = A/4$ (positive, since $A > 0$)

For $S = 0$ singlet, $H = A(s_1 * s_2) = -3A/4$ (negative, since $A > 0$)

The energy difference between triplets (higher energy) and singlet is: $A/4 - (-3A/4) = A$.

Starting from scratch, a positronium could have those “physical” states. The world “physical” here means it is reasonably to imagine. We use $\uparrow$ and $\downarrow$ to stand for spin up and spin down. We have 4 possible “physical” states: both up ($\uparrow\uparrow$), both down ($\downarrow\downarrow$), electron up and positron down ($\uparrow\downarrow$), electron down and positron up ($\downarrow\uparrow$), we here label them:

$$|\uparrow\uparrow\rangle, \quad |\downarrow\downarrow\rangle, \quad |\uparrow\downarrow\rangle \text{ and } |\downarrow\uparrow\rangle$$

Remember, the Hamiltonian H is energy, so we are measuring energy eigen states, not exactly the “physical” states. However, we can represent the energy eigen states by the “physical states”.

Now we use $|S, m_z\rangle$ to stand for the 4 quantum stats of positronium. (Here I omitted all fancy mathematics to be applied, only the results are listed)

Para-Positronium (p-Ps), singlet $S = 0$, $m_z = 0$

$$|0, 0\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$$

Ortho-Positronium (o-Ps), triplets, $S = 1$, $m_z = 1, 0, -1$

$$|1, 1\rangle = |\uparrow\uparrow\rangle$$

$$|1, 0\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$$

$$|1, -1\rangle = |\downarrow\downarrow\rangle$$

Transitions between $|1, 0\rangle$ to $|0, 0\rangle$ are forbidden because that violate 2 conservation laws: Parity Conservation and Angular momentum Conservation.

Zeeman mixing

Now consider a magnetic field along z-direction B (vector), using perturbation method, adding another term to the original Hamiltonian

$$H_B = -\mu_1 \cdot B - \mu_2 \cdot B$$

(dot products)

Both μ_1 and μ_2 are magnetic dipole moment, which is related to electric charge, mass, and spin.

For electron

$$\mu_1 = -g \frac{e}{2m} s_1$$

For positron

$$\mu_2 = g \frac{e}{2m} s_2$$

(Note they have different signs because electrons have negative charge.)

Re-write the perturbation Hamiltonian:

$$H_B = \frac{ge}{2m} B (s_{1z} - s_{2z})$$

(because B is at z-direction, so project to z)

The term $\frac{ge}{2m} B$ is also called Lamore-like frequency, which can be denoted by ω_L

$$H_B = \omega_L (s_{1z} - s_{2z})$$

Now we determine the Zeeman mixing by applying the perturbation H_B to the 4 eigen states of Ps (in which 3 for o-Ps and 1 for p-Ps)

$$H_B |1, 1\rangle = \omega_L (s_{1z} - s_{2z}) |\uparrow\uparrow\rangle = \omega_L \left(\frac{1}{2} - \frac{1}{2} \right) = 0$$

(both spin up)

$$H_B|1, -1\rangle = \omega_L(s_{1z} - s_{2z})|\downarrow\downarrow\rangle = \omega_L\left(-\frac{1}{2} - \left(-\frac{1}{2}\right)\right) = 0$$

So, the $|1, 1\rangle$ and $|1, -1\rangle$ of o-Ps are not influenced by the magnetic perturbation.

Now calculate the perturbation to $m_z = 0$ for o-Ps, which is $|1, 0\rangle$

$$\begin{aligned} H_B|1, 0\rangle &= \omega_L(s_{1z} - s_{2z})\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle) \\ &= \frac{\omega_L}{\sqrt{2}}\left[\left(\frac{1}{2} - \left(-\frac{1}{2}\right)\right)|\uparrow\downarrow\rangle + \left(-\frac{1}{2} - \frac{1}{2}\right)|\downarrow\uparrow\rangle\right] \\ &= \omega_L\left[\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)\right] \\ &= \omega_L|0, 0\rangle \end{aligned}$$

What a surprise. Put Magnetic perturbation onto the $m_z = 0$ triplet state, can reach singlet state $|0, 0\rangle$.

Similarly, applying the perturbation onto the singlet state $|0, 0\rangle$

$$\begin{aligned} H_B|0, 0\rangle &= \omega_L(s_{1z} - s_{2z})\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle) \\ &= \frac{\omega_L}{\sqrt{2}}\left[\left(\frac{1}{2} - \left(-\frac{1}{2}\right)\right)|\uparrow\downarrow\rangle - \left(-\frac{1}{2} - \frac{1}{2}\right)|\downarrow\uparrow\rangle\right] \\ &= \omega_L|1, 0\rangle \end{aligned}$$

Again, perturb the singlet state p-Ps, can reach the triplet $m_z = 0$ state.

How to understand the results that magnetic perturbation acting on the $m_z = 0$ triplet state and the singlet state? The answer is: the perturbation operator H_B couples $|1, 0\rangle$ directly to $|0, 0\rangle$.

Under H_B , transition between triplet $m_z = 0$ state and singlet state are allowed. Remember it is forbidden if there is no magnetic field because that will violate conservation laws of Parity and Angular momentum. With the magnetic field, the angular momentum changed, and so is the Parity of the system. That is, by absorbing or emitting a hyperfine-split (HFS) photon (the parameter A in spin Hamiltonian) of energy 203.4GHz or 8.41×10^{-4} eV.

Note, while implementing a magnetic field B , the HFS remains unchanged.

Options on Magnetic Field – how strong is enough

Given the Quantum physics principle of Zeeman mixing or Magnetic quenching, an important question one will ask is: how strong is the magnetic field needed? In Magnetic Resonance Imaging (MRI), the magnetic field can be as much as 1-3T (Tesla) because the sensitivity is directly related to the magnetic field. To achieve such a high magnetic field, a liquid-nitrogen driven cryogenic system for superconductors must be applied, and this is the major cost for the MR unit.

By passing the cryogenic system seems to be an option to reduce costs. Permanent magnet of 0.5T or slightly higher is commercially available at a much cheaper price; where Zeeman effect is still reasonably distinctive at 0.1T, so the idea could be using permanent magnet (0.1-1T). However, we need to determine the lower limit.

Since the perturbation operator H_B (also called Zeeman operator) only couples triplet $|1, 0\rangle$ and singlet $|0, 0\rangle$ states, we can simplify the matrix into 2x2:

$$\begin{bmatrix} A/4 & 2\mu_B B \\ 2\mu_B B & -3A/4 \end{bmatrix}$$

where A is the superfine split, as mentioned above. Note this matrix is not diagonal, which means $|1, 0\rangle$ and $|0, 0\rangle$ are not the eigen states of the Zeeman operator H_B .

If $2\mu_B B \ll A$, the off-diagonal mixing is small, the matrix is close to diagonal, which means the eigen states are almost pure $|1, 0\rangle$ and $|0, 0\rangle$. Think about $B = 0$, no external magnetic field, then Zeeman perturbation $H_B = 0$, the matrix is diagonal.

Let x stand for the ratio of Zeeman coupling to the hyperfine split (HFS), so

$$x = \frac{2\mu_B B}{A}$$

we know $\mu_B = 5.788 * 10^{-5}$ eV/T; and $A = 8.41 * 10^{-4}$ eV (203.4 GHz)

For a 0.5T permanent magnet, $x \approx 0.069 \approx 0.07$, which is still weak coupling. This means the eigen state is:

$$|\psi\rangle = 0.998|1, 0\rangle + 0.07|0, 0\rangle$$

Please note that for normalization, total probability $0.998^2 + 0.07^2 = 1$, therefore, there are 99.6% probability triplet state, and 0.5% probability of singlet.

Under this situation, $B=0.5T$, the ratio of o-PSs to p-PSs in the coupling is about $99.6/0.5 = 199$. However, the lifetime of p-PSs is 125ps, and that for o-PSs is 142ns. The ratio is $142/0.125 = 1136$. An o-PSs lives 1136 times longer than a p-PSs.

Now let's consider the real situation: originally there are 1,000 o-PSs of $m_z = 0$, and the perturbation B is suddenly applied such that Zeeman mixing occurs. Due to the formula above, right now there are 5 p-PSs and 995 o-PSs (total 1,000). The 5 p-PSs decay quickly in 0.125ns, such that eigen states rebalance instantaneously, due to the same probability, now there are 990 o-PSs and 5 p-PSs (total 995 now). The 5 new p-PSs continuously decay, and so on. Remember the o-PSs will also decay in 142ns, which is 1136 times longer than p-PSs. However, after about 200 times of rebalancing, which is about $0.125ns * 200 = 25ns$, all o-PSs are converted to p-PSs. In another word, 0.5T magnetic field is more than enough.

The limit situation is about 0.1% probability of mixing, which means:

$$|\psi\rangle = 0.9995|1, 0\rangle + 0.032|0, 0\rangle$$

and

$$x = \frac{2\mu_B B}{A} = 0.032, \quad \text{so} \quad B = \frac{Ax}{2\mu_B} = 0.23 \text{ (T)}$$

If B is below 0.23T, not all o-PSs can convert to p-PSs with Zeeman mixing.

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